

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: DSIP (01CT1513)	Aim: To perform audio convolution and inverse filtering using different impulses	
Open-ended-Problem: 02	Date:16-07-2026	Enrollment No:92400133121

Aim:

Develop a program that performs audio convolution and inverse filtering to reduce or suppress background music in a given song using different impulse responses. Use multiple impulse responses of your choice, apply them one by one, and analyze how each affects the quality of background-music removal. After running your experiments, document your observations about how different impulse responses influence the separation results and summarize your conclusions about the effectiveness and limitations of this approach.

Code:

```

from pydub import AudioSegment
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import convolve

audio = AudioSegment.from_mp3("audio.mp3")

audio = audio.set_channels(1)
samples = np.array(audio.get_array_of_samples()).astype(np.float32)

samples_norm = samples / np.max(np.abs(samples))

echo = np.zeros(5000, dtype=np.float32)
echo[0] = 1
echo[2000] = 0.4

inverse_echo = np.zeros(12000, dtype=np.float32)
inverse_echo[0] = 1
inverse_echo[2000] = -0.4
inverse_echo[4000] = 0.16
inverse_echo[6000] = -0.064
inverse_echo[8000] = 0.0256

impulse_responses = {
    "Smoothing": {
        "kernel": np.array([0.25, 0.5, 0.25], dtype=np.float32),
        "inverse": np.array([1, -0.5, 0.25], dtype=np.float32)
    },
    "Strong_Filtering": {
        "kernel": np.array([1, -0.5], dtype=np.float32),
        "inverse": np.array([1, 0.5], dtype=np.float32)
    }
}

```

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```

    },

    "Echo": {
        "kernel":echo,
        "inverse":inverse_echo
    },
}

start = 1000
end = 11000

plt.figure(figsize=(20,10))
plot_no = 1

# Perform convolution
for name, data in impulse_responses.items():

    kernel = data["kernel"]
    inverse_kernel = data["inverse"]

    print("Processing:", name)

    convoluted = convolve(samples, kernel, mode="same")
    # Normalize
    convoluted = convoluted / np.max(np.abs(convoluted))
    convoluted_int16 = (convoluted * 32767).astype(np.int16)

    # Create audio
    convoluted_audio = AudioSegment(
        convoluted_int16.tobytes(),
        frame_rate=audio.frame_rate,
        sample_width=2,
        channels=1
    )

    convoluted_audio.export(f'{name}.wav', format="wav")

    restored = convolve(convoluted,inverse_kernel,mode='same')
    #Normalize
    restored = restored / np.max(np.abs(restored))
    restored_int16 = (restored * 32767).astype(np.int16)

```

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```
restored_audio = AudioSegment(
    restored_int16.tobytes(),
    frame_rate=audio.frame_rate,
    sample_width=2,
    channels=1
)
```

```
restored_audio.export(f'{name}_inverse.wav',format="wav")
```

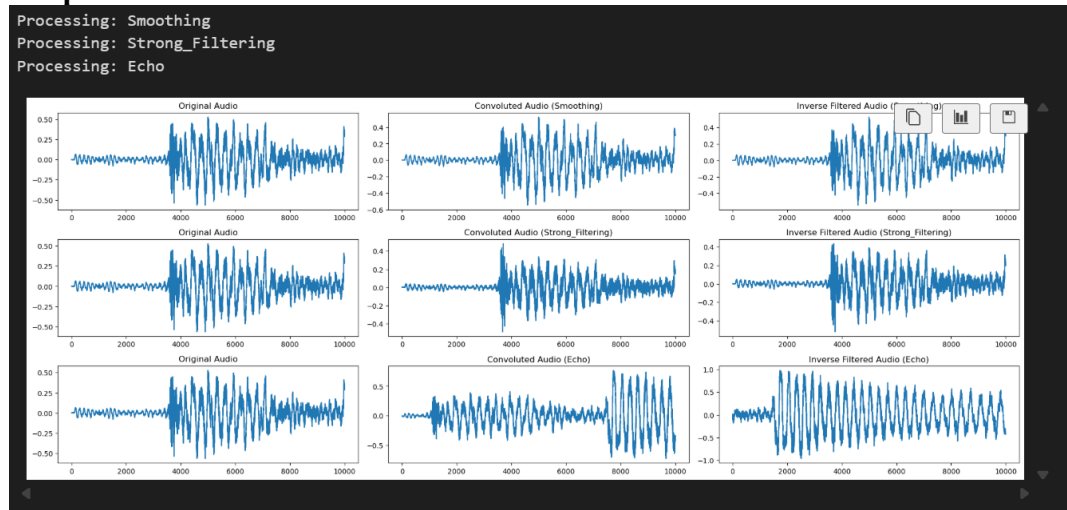
```
plt.subplot(4,3,plot_no)
plt.plot(samples_norm[start:end])
plt.title("Original Audio")
plot_no +=1
```

```
plt.subplot(4,3,plot_no)
plt.plot(convoluted[start:end])
plt.title(f'Convolved Audio ({name})')
plot_no +=1
```

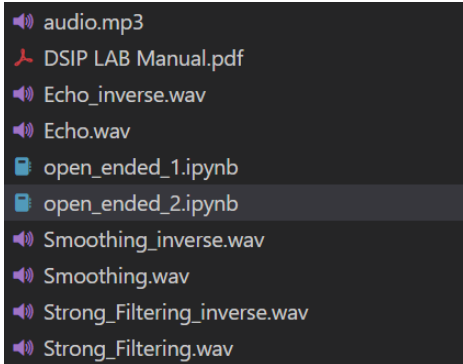
```
plt.subplot(4,3,plot_no)
plt.plot(restored[start:end])
plt.title(f'Inverse Filtered Audio ({name})')
plot_no +=1
```

```
plt.tight_layout()
plt.show()
```

Output:



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Observation:

Smoothing Filter: In this case the vocals are softened and the background music gives a little bit resonant effect than clear echo.

Strong Filter: In this case the vocals are prominent than the background music.

Echo Filter: In this case we can notice proper echo effect, the vocals become more resonant.

→Comparatively when inverse filter is applied echo brings output which is close to the original audio.

Conclusion:

- Through convolution we can observe different impulse responses affecting the audio signal in different ways.
- Depending upon the filter the signal can be echoed or smoothed.
- Inverse filtering gives the partially recovered original signals